

Effort and Weakness in Pair-Dating: A Signaling Model of Need-Display

Diego Rincón

Independent

Abstract

In pair-dating, visible effort functions not as a quality signal but as a *need signal*: it reveals the sender's dependence on the interaction's success. High need reduces desirability. The individually rational response is effort suppression; the collective result is reduced pair formation — a coordination failure structurally analogous to a prisoner's dilemma. We formalize this in a simple signaling game, prove the unique symmetric equilibrium involves effort below the social optimum, and identify two conditions under which effort *increases* desirability (surplus signaling, scarcity framing). The Lacanian structure of desire — desire as desire of the Other, desirability as apparent non-need — is the psychological mechanism behind the equilibrium. The paper describes a real structure. It does not recommend exploiting it.

1 The Signaling Problem

Let two agents A and B interact in a pair-dating context. Each agent i has a hidden *surplus* $s_i \in [0, 1]$: the degree to which they can invest in the other without requiring reciprocation to remain functional. A high-surplus agent can afford the interaction failing; a low-surplus agent cannot.

Each agent chooses a visible *effort level* $e_i \in [0, 1]$. Effort is observable; surplus is not. The receiver j observes e_i and forms a posterior belief $\hat{s}_i(e_i) = \mathbb{E}[s_i \mid e_i]$ about the sender's surplus.

Definition 1 (Desirability). The *desirability* of agent i to agent j is

$$D_j(i) = \hat{s}_i(e_i) - \lambda \cdot e_i,$$

where $\lambda \geq 0$ is the *effort tax*: the degree to which observed effort, beyond its information content, signals demand for reciprocation.

Definition 2 (Effort valence). The *valence* of effort is the sign of $\frac{\partial D_j(i)}{\partial e_i}$:

- *Positive valence*: effort signals surplus ($\hat{s}_i$ increases in e_i).
- *Negative valence*: effort signals need ($\hat{s}_i$ decreases in e_i , or increases slower than λ).

2 When Effort Signals Need

Proposition 3 (The need-display condition). *Suppose that, in the receiver's prior, the distribution of effort satisfies $\text{Cov}(e_i, s_i) < \lambda$. Then effort has negative valence: increasing e_i decreases $D_j(i)$.*

Proof. By Bayes' rule, for s_i, e_i jointly distributed with linear conditional mean, $\hat{s}_i(e_i) = \mu_s + \frac{\text{Cov}(s_i, e_i)}{\text{Var}(e_i)}(e_i - \mu_e)$. Hence $\frac{\partial \hat{s}_i}{\partial e_i} = \frac{\text{Cov}(s_i, e_i)}{\text{Var}(e_i)}$. Then $\frac{\partial D_j(i)}{\partial e_i} = \frac{\text{Cov}(s_i, e_i)}{\text{Var}(e_i)} - \lambda < 0$ under the stated condition. $\square$

Remark 4. The condition $\text{Cov}(e_i, s_i) < \lambda$ holds in romantic contexts for structural reasons: (i) agents with high need exert disproportionate effort to compensate for low surplus (*effort as compensation*), pulling the covariance negative or small; (ii) the effort tax λ reflects the receiver's awareness that effort implies demand, regardless of surplus. Neither condition is irrational. Both are accurate.

3 The Equilibrium

Theorem 5 (Effort suppression equilibrium). *Under the need-display condition (Proposition 3), the unique symmetric Bayes-Nash equilibrium satisfies $e^* < e_{\text{social}}$, where e_{social} is the socially optimal effort level maximizing joint welfare $D_A(B) + D_B(A) - c_A e_A - c_B e_B$.*

Proof. Agent i maximizes their probability of pairing, which is increasing in $D_j(i)$. Under negative valence, $\frac{\partial D_j(i)}{\partial e_i} < 0$, so each agent reduces effort to zero (or to the minimum consistent with maintaining the interaction's existence, call it $e_{\min} \geq 0$). In a symmetric equilibrium both agents play $e^* = e_{\min}$.

The social optimum e_{social} is determined by the condition $\frac{\partial}{\partial e_i}[D_j(i) + D_i(j)] = 0$, which balances the negative signaling effect against the intrinsic value of effort (cooperation, investment, attention). Since the private optimization ignores the externality on $D_i(j)$ (my suppressed effort also reduces the *information* j has about my type, impoverishing the interaction), $e^* < e_{\text{social}}$. $\square$

Corollary 6 (Pair-formation tragedy). *Mutual effort suppression reduces the probability of pair formation below the cooperative optimum. The equilibrium is Pareto-dominated.*

Remark 7. This is a coordination failure, not a moral failing. Both agents act rationally given their beliefs. The structure traps them below the optimum.

4 The Lacanian Mechanism

The signaling model captures the strategic surface. The deeper mechanism is structural: *desire is the desire of the Other*.

In Lacan's formulation, desire is not a drive toward a fixed object but a relation to lack [2]. What is desired is what appears to desire something else — what does not need the desirer. An agent who visibly needs the interaction to succeed has closed the gap: they are already captured. Captured objects are possessed, not desired.

Proposition 8 (Desire and apparent non-need). *Let $\mathcal{D}_j(i)$ denote the desire of j for i , modeled as a function of the perceived gap $g_j(i) = s_j^* - \hat{s}_i(e_i)$, where s_j^* is j 's threshold surplus for engagement. Then $\mathcal{D}_j(i)$ is increasing in $g_j(i)$. In particular: if $\hat{s}_i(e_i) \geq s_j^*$ (agent i appears to need the interaction at most as much as j requires), desire is maintained. If $\hat{s}_i(e_i) < s_j^*$ (agent i appears to need the interaction more than j can supply), desire collapses.*

Remark 9. Proposition 8 is not a formal theorem in the mathematical sense; it is a structural claim about the phenomenology of desire, formalized. The prediction is: the moment the desirer believes themselves desired *out of need* rather than out of surplus, desire extinguishes. This is the mechanism behind the effort tax λ .

5 When Effort Increases Desirability

Effort suppression is not always optimal. Two conditions reverse the valence.

Proposition 10 (Surplus signaling). *If effort is demonstrably costly (the receiver can verify that the sender had better alternatives and chose this interaction), then $\text{Cov}(e_i, s_i) > \lambda$ and effort has positive valence.*

Proof. Costly-signaling logic (Spence [1]): if only high-surplus agents can afford the effort (separating equilibrium), then $e_i > 0$ credibly signals s_i high. The covariance in the receiver's posterior is positive and large. $\square$

Proposition 11 (Scarcity framing). *If the receiver believes the sender has outside options (i.e., other potential pairs with positive probability), effort signals investment rather than need. In this case λ is reduced: effort does not imply demand for reciprocation because the sender could have directed effort elsewhere.*

Remark 12. Propositions 10 and 11 identify the conditions under which effort is not self-undermining. In practice, these conditions are rarely both satisfied simultaneously in early-stage dating, which is why the need-display equilibrium (Theorem 5) is the generic case.

6 The Dark Structure

The equilibrium admits an asymmetric exploitation.

Proposition 13 (Asymmetric suppression). *If agent A suppresses effort while agent B continues to display effort, then: $D_A(B)$ increases (B appears more desirable), and $D_B(A)$ decreases (A appears less needy, hence more desirable by Proposition 8). Both effects favor A .*

Remark 14 (The dark psychology). Proposition 13 is the structural basis of manipulative dating strategies: deliberate effort suppression combined with intermittent reinforcement (occasional high-effort displays that reset the receiver's baseline) maximizes the asymmetric

advantage. The receiver's effort escalates (response to perceived scarcity), the manipulator's apparent desirability increases, and the receiver's actual leverage decreases.

This is described here as a structural feature, not a recommendation. The strategy requires treating the other as a strategic object. It works, on the model's terms, and it damages the person it targets. Stating this directly is the purpose of the paper.

7 Resolution

Theorem 15 (Escape from equilibrium). *The coordination failure of Theorem 5 is resolved if and only if at least one of the following holds:*

- (i) *Effort can be contextualized: the receiver has enough information to distinguish surplus-effort from need-effort.*
- (ii) *The pair exits the signaling regime: communication becomes direct rather than performed, collapsing the gap between e_i and s_i .*
- (iii) *One agent unilaterally accepts the vulnerability of visible effort at the cost of lower short-term desirability, creating an honest signal.*

In cases (i) and (ii), the effort tax λ is eliminated. In case (iii), the equilibrium is broken by a dominated strategy that, if reciprocated, restores the cooperative optimum.

Remark 16. Case (iii) is what is ordinarily called honesty in relationships. It is a dominated strategy in the game-theoretic sense: it lowers expected desirability given a hostile prior. It is the only strategy that makes pair formation genuinely possible.

References

- [1] Spence, M. (1973). Job market signaling. *Quart. J. Econ.* 87, 355–374.
- [2] Lacan, J. (1966). *Écrits*. Éditions du Seuil, Paris.
- [3] Zahavi, A. (1975). Mate selection — a selection for a handicap. *J. Theoret. Biol.* 53, 205–214.
- [4] Crawford, V., Sobel, J. (1982). Strategic information transmission. *Econometrica* 50, 1431–1451.
- [5] Cho, I.-K., Kreps, D. (1987). Signaling games and stable equilibria. *Quart. J. Econ.* 102, 179–221.